Zackary Blaylock

Senior Lead Software Engineer

PROFESSIONAL SUMMARY

Senior lead software engineer with 8+ years building secure, cloud-native systems on AWS. Designed and built four production systems at SendSafely: the Actions workflow engine, an MCP server that connects AI assistants such as Claude to encrypted files, a standalone OAuth 2.1 authorization server, and an AI-driven Custom Form Builder on Amazon Bedrock. Previously built payment and eCredit redemption services at Delta Air Lines. Full-stack across TypeScript, Java, Rust, React, and Angular, with a focus on security, end-to-end encryption, and serverless architecture.

RELEVANT WORK EXPERIENCE

SendSafely, Remote

End-to-end encrypted secure file transfer platform

Senior Lead Software Engineer 12/2020 – Present

- Lead full-stack development across web applications, client SDKs, and third-party integrations built on end-to-end encryption (E2EE) principles.
- Designed and built SendSafely Actions as the sole engineer: an event-driven, serverless workflow engine on AWS Lambda that moves event processing off the core web application and lets customers plug in their own processing, such as DLP and antivirus scanning, through HMAC-signed webhooks. Supports org-wide and personal workflows executed in priority order, with block, hold, release, delete, and deny actions, filter-based branching on earlier step results, and a synchronous pre-finalize deny hook, running in production across US, EU, and AU regions.
- Initiated and built SendSafely's MCP integration, now live for customers, which lets AI assistants such as Claude list, inspect, and manage encrypted packages and workspaces. Built both a local stdio server (36 tools) that handles encryption on the user's machine, keeps Trusted Device keys in the OS keychain, and uses MCP elicitation to get approval before destructive actions, and a stateless remote connector (21 tools) on AWS Lambda that only sees metadata, so file contents and keys never leave the client.
- Architected and built SendSafely's OAuth 2.1 authorization server as a standalone microservice (Koa on Lambda, Aurora MySQL, Angular consent UI) with PKCE, resource indicators, token exchange, rotating refresh tokens with family-wide reuse detection, and RFC 9068 access tokens signed with KMS RSA keys; integrated bearer-token validation, revocation, and per-endpoint scope enforcement into the platform REST API, added an HMAC-authenticated service-to-service layer for deactivation and connection management, and configured the client applications that connect to it.
- Designed and deployed the Custom Form Builder, an AI assistant that turns a chat description into a hosted secure intake form: built on Amazon Bedrock Agents and now moving to Bedrock AgentCore with Claude Sonnet 4.5, a Bedrock Knowledge Base on S3 Vectors, and an MCP gateway, streamed to the browser over SSE with release-pinned agent endpoints and tenant-isolated previews.
- Provision and maintain AWS infrastructure with CloudFormation across ECS, Lambda, EC2, RDS, DynamoDB, S3, and CloudFront, secured with IAM, WAF, and Secrets Manager and monitored with

CONTACT

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SKILLS

Languages:
TypeScript, JavaScript, Java, Python, Rust, SQL, Bash

Frontend:
React, Next.js, Angular, Vue, Redux, Tailwind CSS, SASS, Astro, WCAG accessibility

Mobile & Desktop:
React Native (Expo), Tauri

Backend:
Node.js, Spring Boot, Spring Security, Spring Data JPA, Hibernate, Koa, Axum (Rust), Prisma

Databases:
PostgreSQL, MySQL, Aurora MySQL, MongoDB, DynamoDB, DocumentDB, Drizzle ORM, Flyway

AWS:
ECS, App Runner, Lambda, EC2, VPC, ALB, RDS, S3, S3 Vectors, CloudFront, Route 53, Bedrock, IAM, WAF, Secrets Manager, KMS, API Gateway, SSM, SES, SNS, EventBridge, DynamoDB Streams, CloudWatch, ECR, SAM, CodeBuild, Amplify

Other Cloud:
Google Cloud (GCP), Firebase, Heroku

DevOps & CI/CD:
Docker, Docker Compose, CloudFormation, GitHub Actions, Jenkins, CodeArtifact

Auth & Security:
OAuth 2.1, PKCE, Token Exchange (RFC 8693), JWT (RFC 9068), JWKS, Auth0, Auth.js, HMAC request signing, E2EE

Payments & Messaging:
Stripe, PayPal, Alipay, Apple Pay, Google Pay, PSD2,

CloudWatch and SNS.

- Built and shipped integrations for Freshdesk, Zendesk, Salesforce, Outlook, Slack, Chrome, and a serverless email gateway, and maintain client API SDKs in Java, Node.js, Python, and .NET.
- Created Claude agent skills, adopted across the engineering team, that standardize development workflows.
- Enhanced CI/CD pipelines with GitHub Actions, CodeArtifact, and AWS deployment strategies, improving release consistency and reducing manual deployment steps.
- Partner with Product Management on technical planning and roadmap execution, and lead triage of customer-reported defects through fix and release.

Delta Air Lines, Atlanta, GA

Full Stack Software Engineer 03/2019 – 12/2020

- Developed and maintained a payment processing engine built on Spring Boot REST APIs and microservices, supporting all major card networks plus PayPal, PayPal Credit, Alipay, Apple Pay, Google Pay, and instant bank transfer, including CID/CVV verification and PSD2 compliance.
- Built the backend for redeeming Delta eCredits, eCertificates, Gift Cards, and other e-documents as payment on delta.com, partnering closely with the frontend and eligibility-rules teams to deliver the feature quickly and reliably (Certificates, eCredits & Gift Cards).
- Led development of an Angular payment portal UI integrating multiple payment methods, ensuring WCAG accessibility compliance.
- Maintained Jenkins and Gradle CI/CD pipelines with JUnit and Jest test coverage, partnering with QA on test automation.
- Provided production support and break-fix response for critical payment defects affecting customer transactions.

PREVIOUS EXPERIENCE

Frontend Software Developer, Cox Automotive, Remote 01/2019 – 03/2019

Software Developer, Hiliteu (Startup), Remote 01/2018 – 01/2019

PROJECTS

SendSafely Actions: Multi-tenant workflow automation for an E2EE file-sharing platform. TypeScript Lambdas build an execution plan for each event (package finalized, workspace file uploaded, recipient added), run native actions through the platform REST API, and pause on HMAC-signed webhooks to customer services (antivirus, DLP, archiving) until they call back, with an EventBridge-scheduled cleanup for 15-minute timeouts, replay of failed runs, and rate-limited SES notifications. Configuration is stored in MySQL (Hibernate), and execution state in DynamoDB, streamed to DocumentDB for customer-facing logs. Deployed with AWS SAM and CodeBuild, with CloudWatch alarms to SNS. Also built the proof-of-concept OPSWAT antivirus webhook receiver.

SendSafely MCP Server: TypeScript MCP server (npm workspaces monorepo) shipped two ways: a published local stdio package with OS-keychain credentials, client-side encryption, and sandboxed downloads, and a stateless Streamable HTTP connector on Lambda behind CloudFront, WAF, and API Gateway that serves RFC 9728 protected-resource metadata, rate-limits per token, and exchanges the user's OAuth token for a tenant-scoped token on every call. Packages are published through GitHub Actions with OIDC to CodeArtifact, and the same tested tarball is promoted to production.

SendSafely OAuth Authorization Server: OAuth 2.1 microservice (Koa on Lambda behind CloudFront, Aurora MySQL with Drizzle ORM, Angular sign-in

Twilio

Testing:
Jest, Vitest, JUnit, Jasmine, Karma

AI:
Amazon Bedrock Agents, Bedrock AgentCore, Bedrock Knowledge Bases, Anthropic API, OpenAI API, OpenRouter, Claude Code, Cursor, MCP

Architecture:
REST APIs, OpenAPI, Server-Sent Events, Microservices, Multi-tenancy

Tools:
Git, GitHub, GitLab, npm, Yarn, Maven, Gradle, Cargo, Vite, Webpack

EDUCATION

University of Kentucky
Bachelor of Business Administration
Management
Lexington, KY
05/2016

Kennesaw State University
Certificate in Computer Science Foundation
Kennesaw, GA
05/2017

Thinkful
Full Stack Web Development Immersion (Bootcamp)
03/2018

and consent app) supporting PKCE, resource indicators, token exchange, introspection, and revocation. Users sign in through the main application via a KMS-signed, single-use handover; tokens and client secrets are hashed at rest; refresh tokens rotate with family revocation and alerts on reuse; and Java servlet filters validate bearer JWTs against JWKS, check revocation through a cached introspection client that fails closed, and enforce scopes on every API endpoint. Blue/green Lambda alias deploys.

SendSafely Custom Form Builder: Chat-driven builder for secure Dropzone intake forms. A Java SSE streaming servlet drives a Bedrock agent (Bedrock Agents in production; a Bedrock AgentCore harness with Claude Sonnet 4.5, session Memory, and an MCP gateway to a Knowledge Base on S3 Vectors with Titan v2 embeddings) that writes a self-contained HTML form, publishes a preview to S3, and serves it with a nonce-based CSP and server-side placeholder substitution. Release-pinned agent endpoints let each app release ship its own prompt and knowledge base, deployed by a scripted CloudFormation pipeline with least-privilege IAM per resource.

Delta eCredits, Certificates & Gift Cards: Backend services (Java, Spring Boot) that let customers apply eCredits, eCertificates, Gift Cards, and other e-documents as payment when booking on delta.com, built alongside the frontend and eligibility-rules teams (Certificates, eCredits & Gift Cards).

Delta Payment Engine: Backend payment services (Java, Spring Boot REST APIs and microservices) for delta.com checkout, supporting all major card networks plus PayPal, PayPal Credit, Alipay, Apple Pay, Google Pay, and instant bank transfer, with CID/CVV verification and PSD2 compliance. Maintained and hardened the engine in production, including break-fix response for critical payment defects.

PQVault *(personal side project, in development):* Post-quantum encrypted password manager with desktop (Rust, Tauri), browser extension, web, and mobile (React Native, Expo) clients built on a shared React/TypeScript UI library. Rust (Axum) API on PostgreSQL with AWS infrastructure defined in CloudFormation. Uses ML-KEM-768 for device keys and native biometric unlock (Touch ID, Windows Hello).